

The vision for the BSA Tool

To support the Sendai Framework guidelines for integrating **disaster risk management** into the Transportation sector

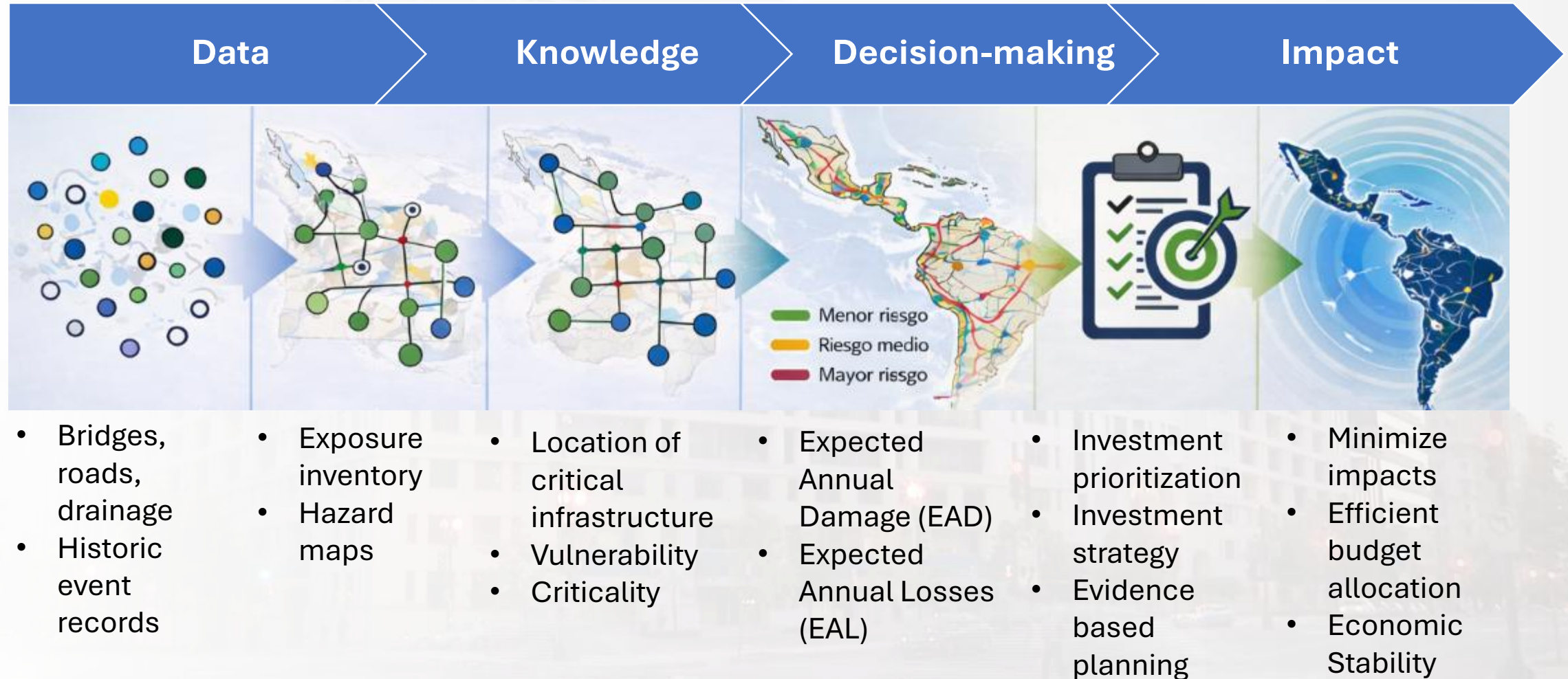
Institutional Commitment:

Providing resources for investment decision-making in resilient public infrastructure.



How does BSA transforms information into impact for the region?

From sparse data to strategic decisions...



BSA helps governments to make evidence-based decisions

What is the Blue Spot Analysis?



Methodology and Tool

To strengthen the planning system



Identify and prioritize investments

Within transportation infrastructure



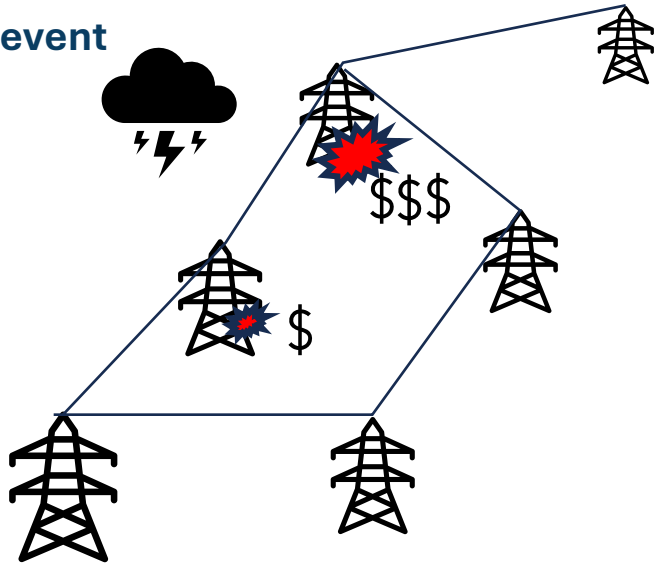
Provide insights

To strengthen the resilience of infrastructure to disaster risk

BSA Key Concept

Difference between damages and losses

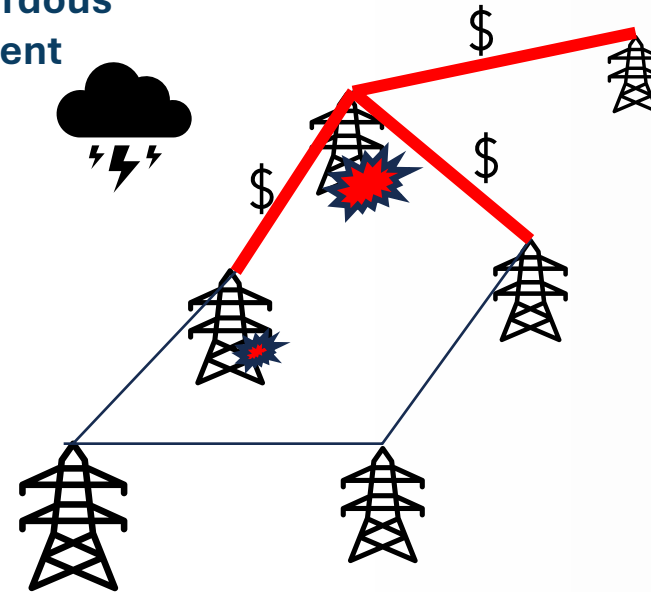
Hazardous event



Damage

(Cost of repairing structures)

Hazardous event



Losses

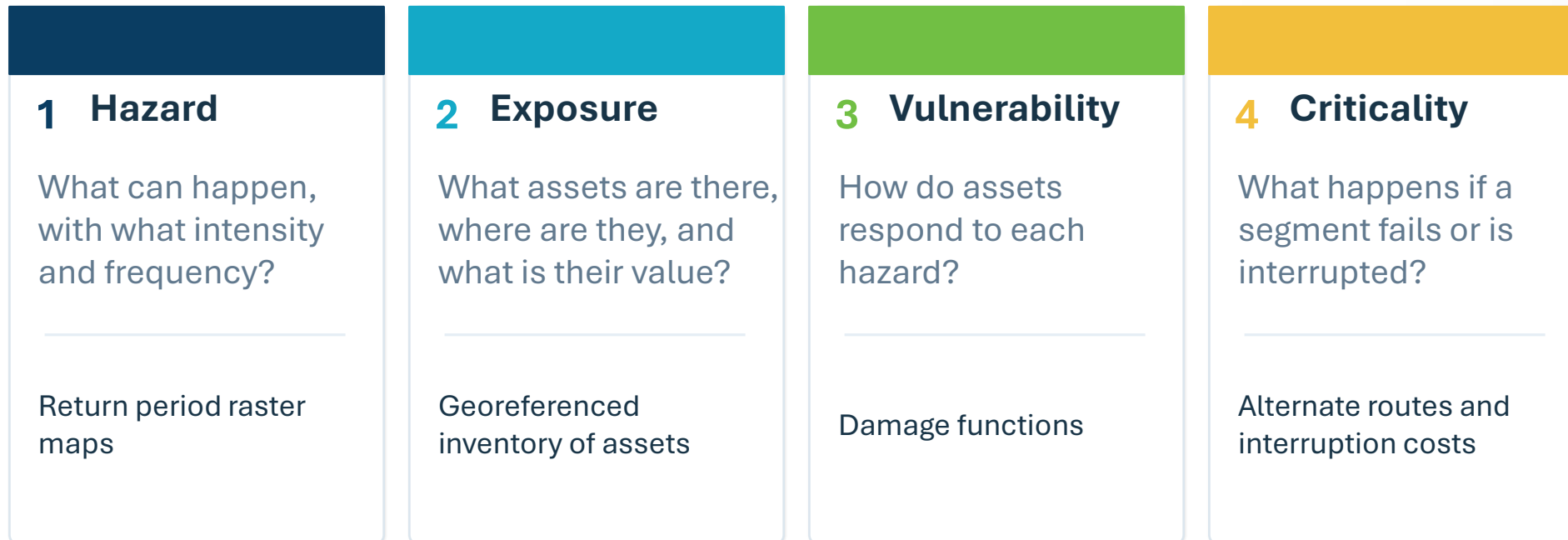
(Cost of service interruption)



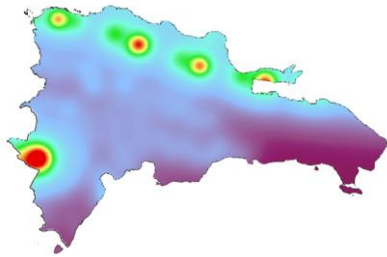
BSA Methodology



Four components that allow to organize the assessment process and transform the data into decision-making insights



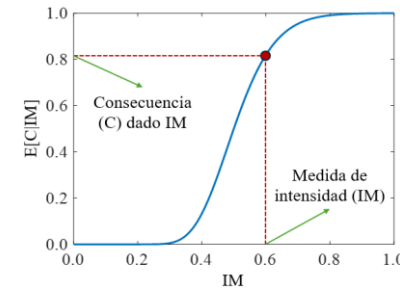
BSA Workflow



Hazard



Exposure



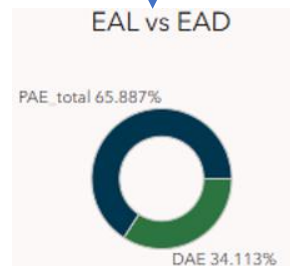
Vulnerability



Criticality



Blue Spot
Analysis
2.0



Statistics



Damage Maps

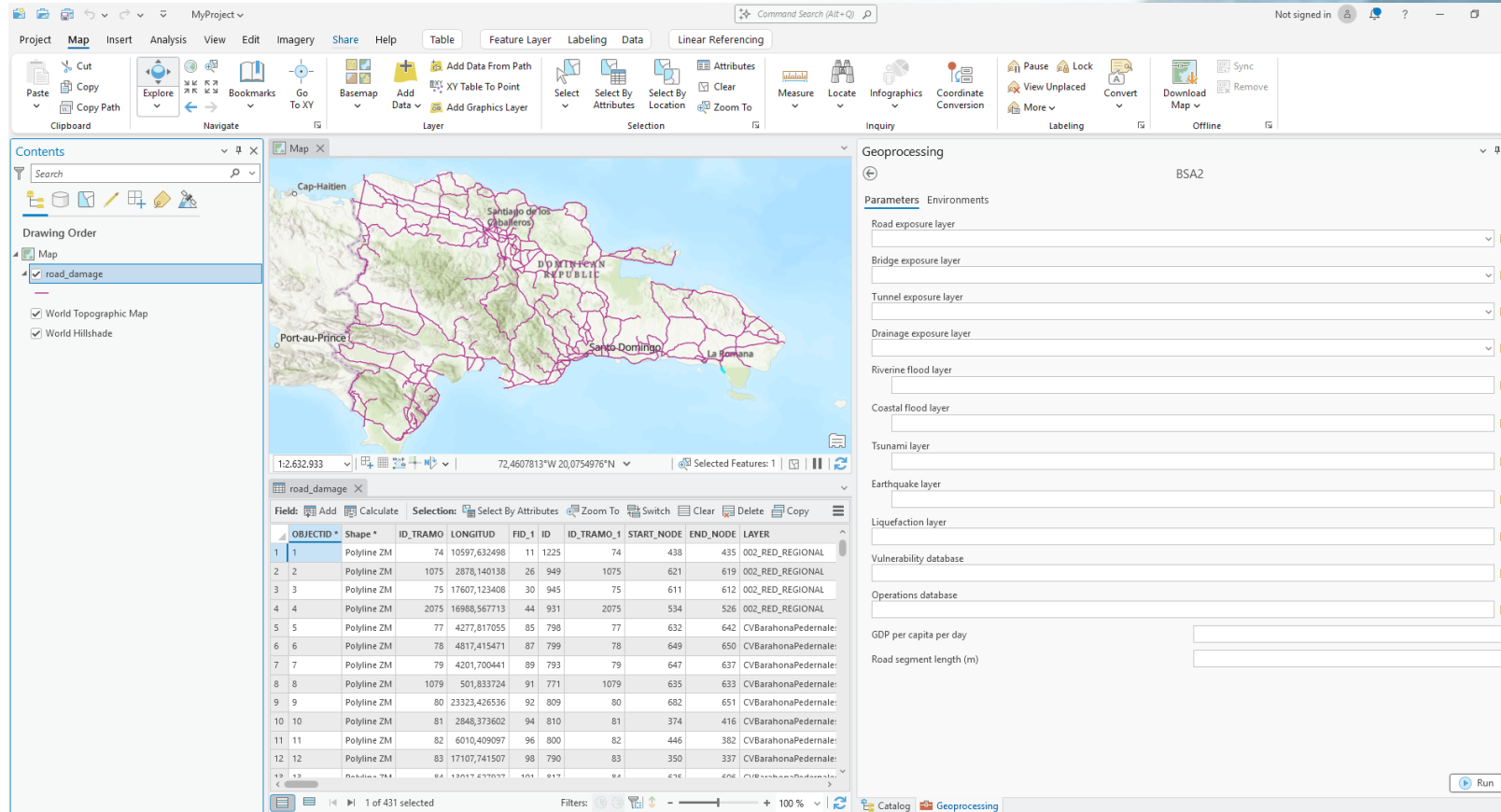


Loss Maps



Prioritization

BSA Desktop Tool (ArcGIS)



Why is it useful?

Modular

Loads asset inventories, hazard layers, vulnerability functions and operating costs.

Reproducible

Standard process to aid in comparison between regions and countries

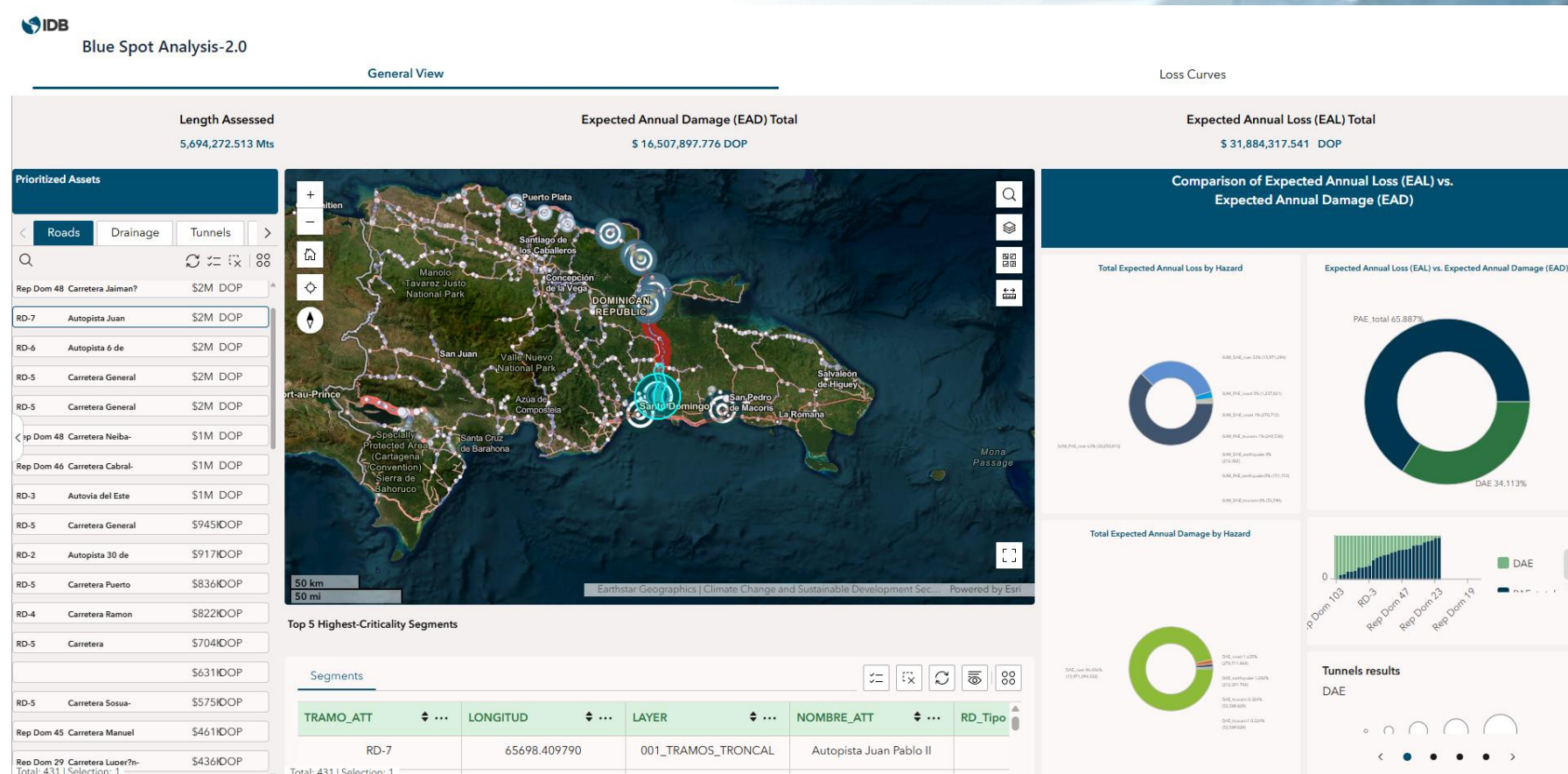
Scalable

Works on transportation networks of different sizes and includes multiple types of assets to analyze

Focused on decision-making

Beyond maps: estimates damages, losses and priority rankings

BSA Online Tool (Dashboard)

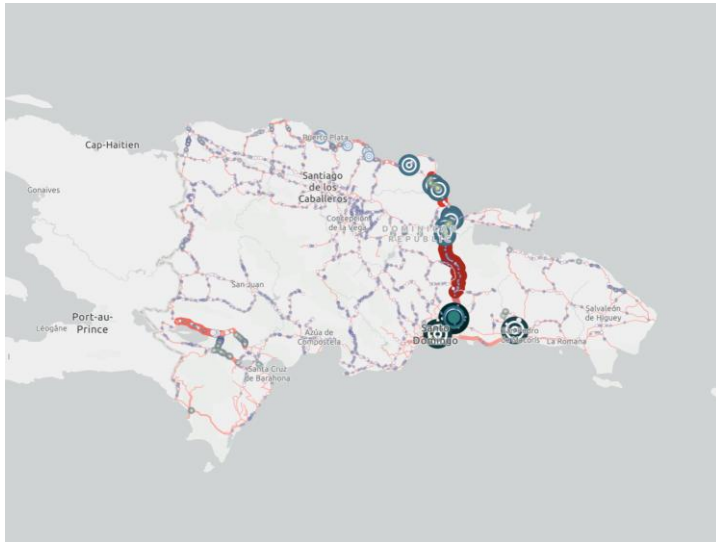


User functionality

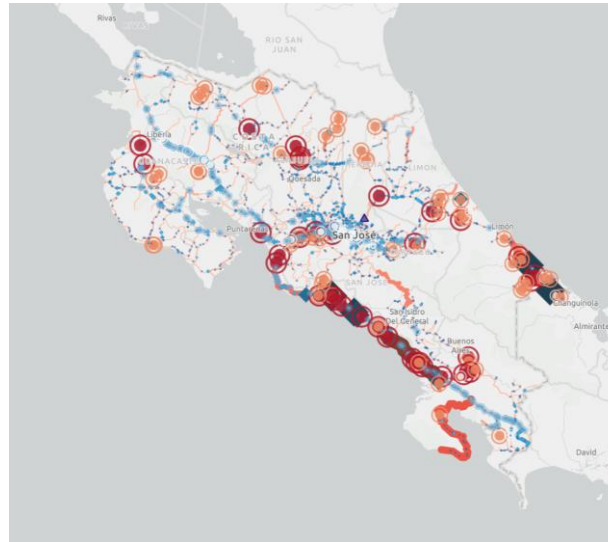
- Priority ranking
- EAL and EAD comparison
- Explore results per hazard
- Prepare reports and communicate to audiences
- Visualize Loss Exceedance Curves

Beyond mapping: it is an interphase to easily visualize and interpret data, which allows a natural conversation with teams and decision-makers

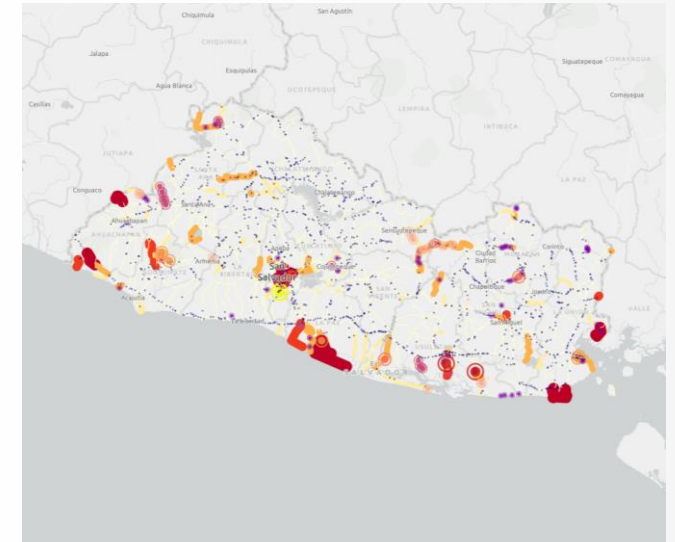
BSA Results (so far...)



Dominican Republic
(Development benchmark)



Costa Rica
(Preliminary)



El Salvador
(Preliminary)

Required datasets

1. Asset inventories

Road network, bridges, tunnels, drainage, location, physical attributes, and replacement costs.

2. Hazard layers

Rasters/maps per hazard and return period (historical + climate change scenarios)

3. Vulnerability and recovery

Damage functions, unit costs, recovery times, and performance parameters.

4. Criticality and operating costs

Average Daily Traffic (AADT), alternative routes, occupancy, operating costs, and daily GDP proxies.

**Download the
input data
fact sheet**



[IDB_InputData_BSA_V1_25022026_1.xlsx](#)